



Firewall Concepts

Most, if not all, of you are familiar with **personal firewalls** or **host-based firewalls** to protect PCs/laptops as shown in Figure 1. However, it can be difficult to manage hundreds or thousands of hosts possibly running different personal firewalls in an enterprise.

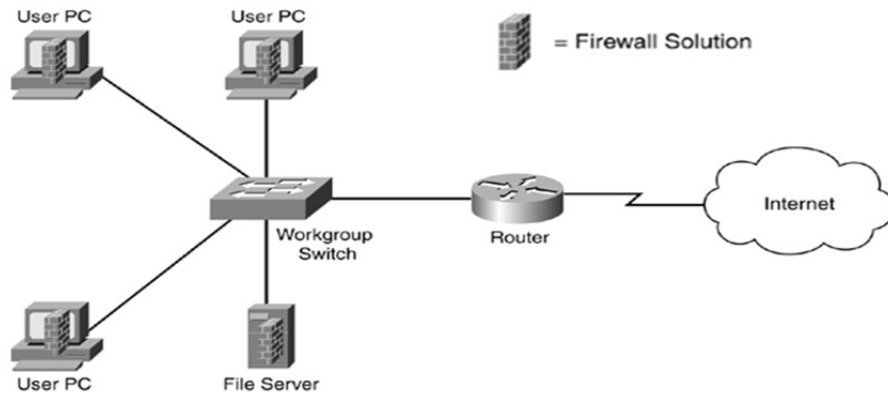


Figure 1

Instead, it is more manageable to deploy one or few **network firewalls** which are typically dedicated appliances implemented at the network edge/perimeter of the enterprise network as shown in Figure 2.

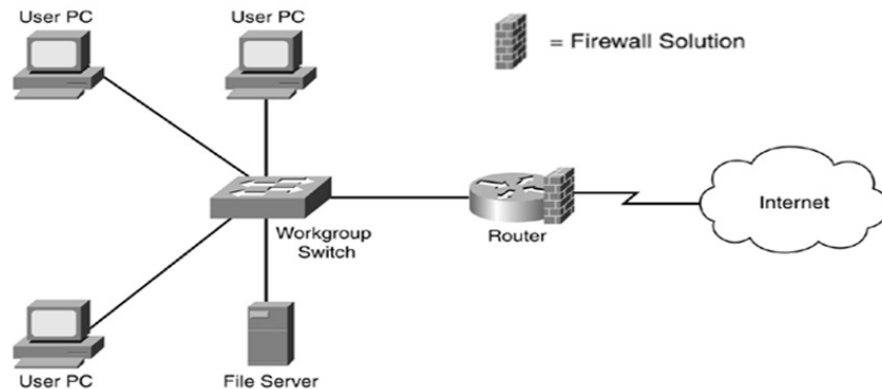


Figure 2

Theoretically, a **firewall** is to segment a network into two or more **security zones**. Figure 3 shows an example of deploying a firewall to segment a network into two security zones - an **internal (protected)** zone, and an **external (untrusted)** zone.

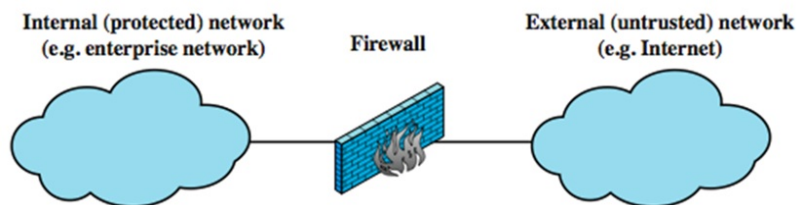


Figure 3

To have more than two security zones, one or more firewalls may be deployed. Figure 4 shows an example of deploying one firewall to segment a third security zone called **demilitarized zone (DMZ)** when the enterprise needs to host servers to be accessible from the Internet.

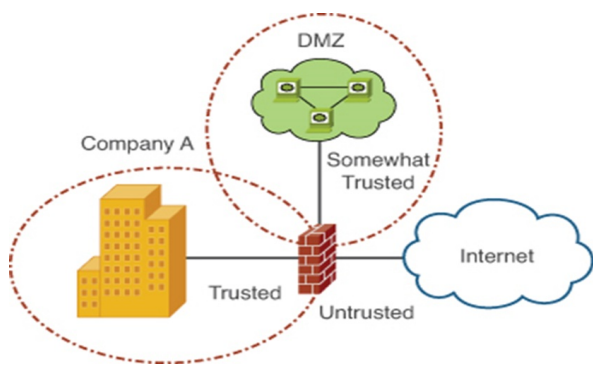


Figure 4

Alternatively, Figure 5 shows an example of deploying two firewalls where the second firewall is used to further segment internal 'trusted' enterprise network into more **trusted** data center network and less untrusted/**untrusted** internal network to defend against users whom may be potential attackers to launch insider attacks to the data center.

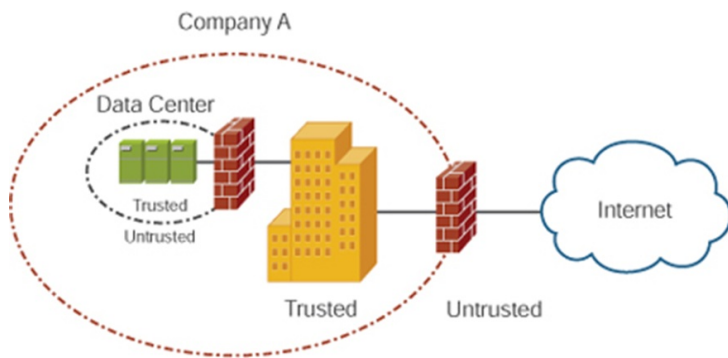


Figure 5

Technically, there are three common types of firewalls which can be found deployed in practice.

The first type is **packet-filtering firewall** which can filter frames/packets based on layer-2 MAC addresses, layer-3 IP addresses, and layer-4 port numbers as shown in Figure 6. In fact, a router configured with ACL is considered a packet-filtering firewall.

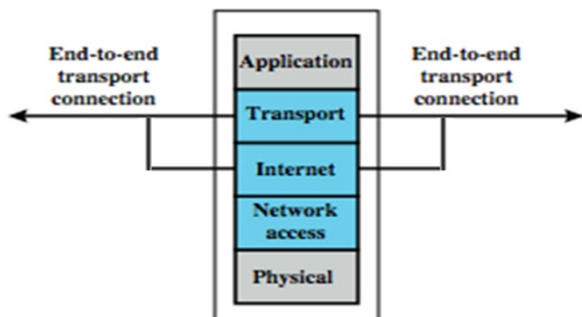


Figure 6

The second type is **stateful firewall** as shown in Figure 7 which operates similarly as packet-filtering firewall but offers better security due to the maintaining of state information. In the industry, the term firewall is synonym with stateful firewall which we will experiment in the lab exercises.

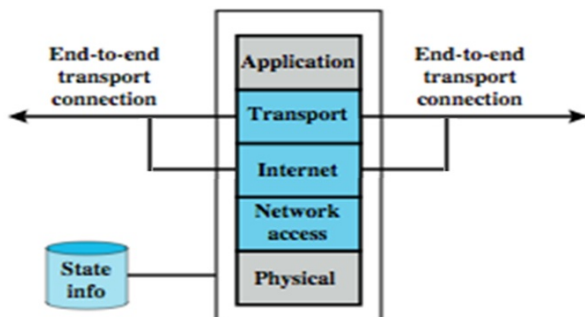


Figure 7

The third type is **application proxy firewall** as shown in Figure 8 which is specially designed to filter a specific application. An example is the Web Application Firewall (WAF) for filtering web traffic.

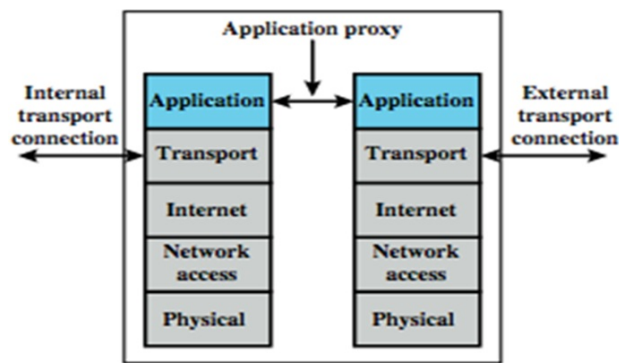


Figure 8

If a company has no budget constraint, all three types of firewalls may be deployed together. Figure 9 shows an example of deploying a **packet-filtering firewall** to first perform fast-filtering of unwanted packets based on ACLs. Next, packets which are permitted will go through **stateful firewall** for further filtering. For added security, **WAF** may also be deployed at the DMZ to further filter web traffic going to the web server.

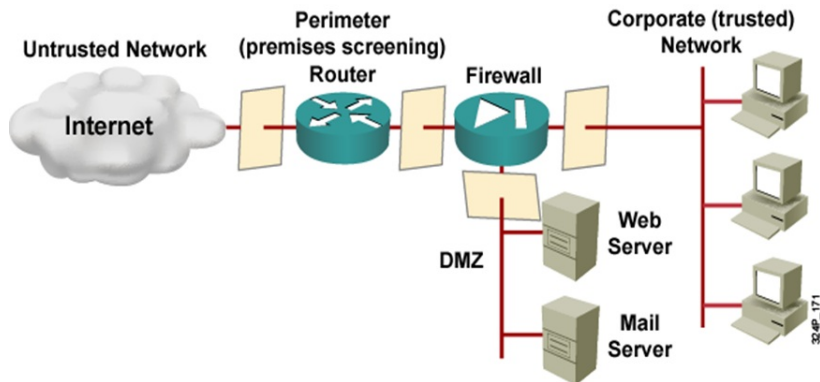


Figure 9

For better understanding, we will next move on to study real-world firewalls.